

MUHAMMAD KAZIM, Ph.D.

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AI for Energy Systems Sustainability | Multilayer Network Researcher

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RESEARCH PROFILE

A results-driven researcher with a Ph.D. in Industrial & Manufacturing Engineering from North Dakota State University (USA), specializing in AI-based multilayer network analysis for smart and sustainable energy systems. My work focuses on graph neural networks (GNNs), predictive maintenance, and resilience modeling of large-scale energy infrastructures. I have a proven track record of developing and applying novel computational frameworks to address complex, real-world challenges in critical infrastructure resilience, cybersecurity, and biomedical AI. My interdisciplinary research has been published in high-impact journals and presented at international conferences, reflecting a commitment to advancing translational research and fostering collaborations with academic and industry partners. Currently serving as Visiting Faculty at the University of Malakand, Pakistan, I aim to advance interdisciplinary research that integrates artificial intelligence, critical infrastructure resilience, and sustainable energy transitions.

Education

- **Doctor of Philosophy (Ph.D.), Industrial & Manufacturing Engineering, North Dakota State University, Fargo, USA, Conferred August 2025.**
 - **Thesis:** Enhancing Resilience in Energy Networks: A Multilayer Network and Machine Learning Approach
 - **Project:** AISustain Project (AI for Resilient Energy Infrastructure) aisustain.com/research
 - **Focus Areas:** Multilayer Networks, Graph Neural Networks, Energy System Vulnerability, Predictive Maintenance.
- **Master of Science (MS), Computer Science, Abasyn University, Peshawar, Pakistan, Conferred August 2019**
 - **Thesis:** Enhanced Detection Mechanism for Application-Layer DDoS Attacks using Machine Learning.
- **Master of Science (M.Sc), Computer Science, University of Malakand, KPK, Pakistan, Conferred March 2014**
 - **Awarded Gold Medal** for securing First Division with distinction.
 - **Focus:** Artificial Intelligence, Software Engineering, and Data Networks.

RESEARCH EXPERTISE

- **AI for Energy Systems:** Multilayer network analysis, Graph Neural Networks for power grids, predictive maintenance, energy resilience and vulnerability assessment, smart grid optimization.
- **Network Science & Graph Analysis:** Multilayer networks, complex systems modeling, link prediction, network resilience, interdependency analysis.

- **Machine Learning & AI:** Graph Neural Networks (GNNs), deep learning architectures, predictive modeling, explainable AI, autonomous systems.
- **Cybersecurity & Digital Forensics:** Deepfake detection, digital forensics, cybersecurity, privacy protection, anonymity analysis.
- **Biomedical AI & Computational Biology:** Reverse vaccinology, protein-protein interaction networks, vaccine candidate discovery, Biomni prioritization, inductive representation learning.

ACADEMIC & PROFESSIONAL EXPERIENCE

I. Visiting Faculty *Dept. of Artificial Intelligence, University of Malakand, Pakistan / September 2025 – Present* - Teaching undergraduate and graduate courses in AI, Data Structures, and Programming. - Supervising research projects in intelligent systems and machine learning. - Contributing to departmental development under the Faculty of Computing & Engineering.

II. Graduate Research Assistant, *Dept. of Industrial & Manufacturing Engineering, NDSU / June 2023 – August 2025 (In collaboration with AISustein)* - Developed AI models for multilayer energy systems using real-world data from the Belgian power infrastructure and seven years of incident data from Oklahoma Gas & Electric Company. - Created predictive maintenance systems using Graph Neural Networks (GNNs) - Conducted research on fault propagation, energy resilience, and interdependencies using advanced machine-learning techniques.

III. Graduate Teaching Assistant, *Dept. of Computer Science, NDSU / Aug 2022 – May 2023* - Led lab sessions for CSCI 172 (Intermediate Visual Basic). - Evaluated student projects and provided technical mentorship.

IV. Data Processing Supervisor (C/O) *BISE Malakand, Pakistan / June 2017 – April 2022* - Automated transcript processing & payroll for 120+ employees - Deployed Oracle PL/SQL & OMR for student evaluation

V. Software Engineering & Systems Development (2014–2017) *LGE&RDD, Pakistan / June 2014 – 2017* - Created government platforms for citizen services & e-bidding systems. - Upgraded web apps using Core PHP, CodeIgniter, Oracle, and AngularJS

FUNDED RESEARCH

1. AISustein: AI for Resilient Energy Infrastructure - Funding Agency: National Science Foundation (NSF) - **Project Website:** <https://www.aisustein.com/about> - **Role:** Graduate Research Assistant / Ph.D. Researcher - **Period:** June 2023 – August 2025 - **Description:** Primary funding source for doctoral research on multilayer energy networks and predictive maintenance. This project provided access to real-world utility data from Oklahoma Gas & Electric Company (OGE) spanning seven years of incident reports, as well as the Belgian power infrastructure dataset. Research outcomes include four peer-reviewed journal publications and one conference paper on GNN-based predictive maintenance, link prediction, and network resilience.

2. Network-Integrated Reverse Vaccinology for Flavobacterium - Funding Agency: U.S. Department of Agriculture (USDA) - **Role:** Graduate Research Assistant - **Period:**

Summer 2025 - **Description:** Developed a novel computational framework integrating a biomedical AI agent (Biomni) with Graph Neural Networks (GNNs) to identify vaccine candidates for *Flavobacterium*, a significant pathogen in aquaculture. This work resulted in a preprint and is currently under peer review.

INDUSTRY AND UTILITY PARTNERSHIPS

- **Oklahoma Gas & Electric Company (OGE):** Collaborated on a comprehensive analysis of seven years of real-world incident data from a major North American utility. This partnership resulted in the development and validation of predictive maintenance models that demonstrated commercial viability and real-world applicability of GNN-based approaches for grid resilience.
- **Research Collaboration:** Demonstrated ability to work with industry partners to access proprietary operational data and translate academic research into practical solutions for critical infrastructure management.
- **Commercialization Pathway:** Committed to translating research into industry-ready solutions through partnerships with utility operators and energy sector stakeholders.

RESEARCH PUBLICATIONS

Journal Articles

1. **Kazim, M.**, Pirim, H., Shi, S., & Wu, D. (2024). Multilayer analysis of energy networks. *Sustainable Energy, Grids and Networks*, 39, 101407. <https://doi.org/10.1016/j.segan.2024.101407>.
2. **Kazim, M.**, Pirim, H., Yadav, O. P., Le, C., & Le, T. (2025). Analysis of multilayer energy networks: A comprehensive literature review. *Applied Energy*, 398, 126357. <https://doi.org/10.1016/j.apenergy.2025.126357>.
3. **Kazim, M.**, Pirim, H., Le, C., Le, T., & Yadav, O. P. (2025). Multilayer GNN for Predictive Maintenance and Clustering in Power Grids. *iScience*.
4. **Kazim, M.**, Pirim, H., Yadav, O. P., Le, C., & Le, T. (2025). Edge-level explainable graph neural networks with network-centric features for transmission line failure prediction in power grids. *Sustainable Energy, Grids and Networks*. <https://doi.org/10.1016/j.segan.2025.101969>.
5. Sohail, S., Sajjad, S. M., Zafar, A., Iqbal, Z., Muhammad, Z., & **Kazim, M.** (2025). Deepfake Image Forensics for Privacy Protection and Authenticity Using Deep Learning. *Information*, 16(4), 270. <https://doi.org/10.3390/info16040270>.
6. Javed, M. S., Sajjad, S. M., Mehmood, D., Mansoor, K., Iqbal, Z., & **Kazim, M.** (2024). Analyzing Tor Browser Artifacts for Enhanced Web Forensics, Anonymity, Cybersecurity, and Privacy in Windows-Based Systems. *Information*, 15(8), 495. <https://doi.org/10.3390/info15080495>

Conference Proceedings

7. **Kazim, M.**, Pirim, H. et al., Link Prediction and Navigability of Multiplex Energy Networks, *ASCE International Conference on Computing in Civil Engineering*, July 28-31, 2024

Under Review

8. **Kazim, M.**, Farzana, F., Pirim, H., Hanson, L., Griffin, M., & et al. (2025). Network-Integrated Reverse Vaccinology Using Biomni-Prioritized Features and Graph Neural Networks in Flavobacterium. *Preprint available at Research Square*. <https://doi.org/10.21203/rs.3.rs-8107499/v1>

RESEARCH PROPOSALS IN PREPARATION

- **Title:** Self-Healing Energy Grids with Autonomous AI Agents
Concept: To develop an autonomous grid management system that integrates a Semantic Energy Knowledge Graph (SEKG), Multilayer GNNs, and autonomous AI agents to predict and prevent cascading failures, and optimize energy flow in real-time.
- **Title:** A Scalable AI-Powered Platform for Rapid Vaccine Discovery
Concept: To create a generalizable, modular platform for rapid vaccine candidate discovery across multiple pathogen systems by integrating multi-omics data with advanced GNN architectures.

TECHNICAL SKILLS

- **Programming Languages:** Python, R, SQL, MATLAB, C++
- **Machine Learning/AI:** PyTorch, TensorFlow, scikit-learn, Keras, GNNs (GCN, GAT, GraphSAGE), Deep Learning, Reinforcement Learning
- **Data Science & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, NetworkX
- **Web Development:** HTML/CSS, JavaScript, PHP, CodeIgniter, AngularJS
- **Databases:** Oracle, MySQL, PostgreSQL
- **Tools & Platforms:** Git, Docker, Jupyter Notebook, LaTeX, Linux